

Kyle Main

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Senior Data Engineer — Large-Scale Pipelines, GenAI/LLM Tooling & Platform Engineering

Data engineer with 12 years building large-scale data pipelines and detection-content platforms, including ground-up ingestion architecture for 220+ heterogeneous data sources, a Common Information Model standardizing schema across all of it, and Apache Beam/GCP Dataflow pipelines for high-volume historical retrieval. Direct, hands-on GenAI/LLM experience applying prompt engineering to production security workflows and orchestrating SIEM APIs via GenAI-powered tooling, plus a full CI/CD pipeline (GitLab) with automated testing, staged/safe rollout, and quality-metrics tracking — the same discipline this role's AI/ML data platforms require. Cybersecurity background includes hands-on ingestion and detection work built directly on CrowdStrike telemetry.

CORE SKILLS

Data Engineering & Pipeline Architecture: Ground-up ingestion pipelines for 220+ heterogeneous log/data sources; Common Information Model design — a data dictionary standardizing field names/types/schema across every source; 50+ Logstash filters for parsing/normalization; data-quality monitoring and alerting; homegrown Apache Beam program on GCP Dataflow for high-volume historical/cold-storage retrieval

Cloud & Distributed Data Processing: GCP (Dataproc, BigQuery, Dataflow); Spark/PySpark/SparkSQL for large-scale distributed processing; GCP serverless/event-driven enrichment; working familiarity with Kafka and Flink (hands-on exposure to a Flink-job environment)

GenAI / LLM Applications & Tooling: Prompt engineering for security use cases — data analysis, false-positive identification, automated detection-content generation; GenAI-driven orchestration of SIEM APIs across 9 platforms; reusable GenAI-powered tooling built for other engineers (e.g., automated cross-SIEM detection-rule conversion)

Multi-Platform Orchestration & CI/CD: Built and run a full CI/CD pipeline (GitLab) for detection/content deployment across 9 platforms via native APIs, with automated unit/integration testing, staged/safe production rollout, formally tracked quality metrics, and multithreaded parallel deployment

Platform & Access Engineering: Python, SQL, Git; Docker (custom images, containerized test environments); worked within a Kubernetes-orchestrated platform as a user; Cloud IAM policy/role implementation (AWS, GCP)

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built an entirely new data ingestion platform from scratch (DOE/NNSA project, completed) pulling CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch environment — custom data transforms, a UEBA detection layer built on top of them, data-quality monitoring/alerting, and custom dashboards.
- Building production GenAI tooling for security automation (current): prompt engineering for detection-content generation and false-positive triage, plus GenAI-driven orchestration of SIEM APIs across multiple platforms.
- Supported data ingestion/quality engineering (CISA CDM at DOE, completed) across a combined Elasticsearch/Splunk environment; currently manages detection/alerting analytics (Splunk saved searches) for Treasury's SOC (TSSOC, current project).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, statistical, behavioral, and ML-based detection content and pipelines against massive-scale customer data on a cloud-based big-data platform.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Owned data engineering for a next-gen cloud-based analytics platform at massive scale — 220+ ingested log sources, 50+ Logstash parsing/normalization filters, and a Common Information Model standardizing field names/types across all of it — as a very early hire building the pipeline from scratch.
- Built a homegrown Apache Beam program run via GCP Dataflow to fetch high-volume historical/cold-storage data on request; owned connector/pipeline health monitoring and troubleshooting.
- Ran exploratory data analysis at scale on GCP Dataproc using Zeppelin notebooks and PySpark/SparkSQL; created/managed 2,300+ detection rules covering most of the MITRE ATT&CK matrix, including ML-based clustering and time-series anomaly detection models.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security log data to build custom detection models — DNS-based malware detection/mitigation and anomalous-behavior discovery across the network.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox · Security Clearances: Top Secret (current), DOE Q (held), Public Trust (held)